Theorem 4.15

"Let U be the universe of discourse. Suppose that for each $x \in U$ we have a statement $P(x)$, which might be true or false depending on x. Then:"

a. We know $\neg[(\forall x \in U)P(x)]$ is equivalent to $(\exists x \in U)(\neg P(x))$.

Proof.

$\implies$ Assume it is not true that for all $x \in U$, $P(x)$ is true. Then there exists at least one x which implies $P(x)$ is false.

$\impliedby$ Assume there exists an $x \in U$ called x', which implies $P(x)$ is false. Then $\forall x$ includes x'. Thus $\forall x$ does not imply $P(x)$ is true.

Thus, the two statements imply each other. Therefore, the original statement is true.

■

Proof.

b. We know $\neg((\exists x \in U)P(x))$ is equivalent to $(\forall x \in U)(\neg P(x))$.

$\implies$ Assume there exists no x such that $P(x)$ is true. Then for $\forall x$, we know $P(x)$ is never true.

$\impliedby$ Assume $\forall x$, then $P(x)$ is always false. Then there exists no x which makes $P(x)$ true.

Thus, the two statements imply each other. Therefore, the original statement is true.

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Theorem 4.23

Proof.

We use proof by contradiction. Thus, assume that $\exists n : 3n^2 + n + 14$ is odd. There are two cases, where n is even or not.

If n is even, then, $n = 2k$ for some integer k. Thus, $3n^2 + n + 14$

$$= 3(2k)^2 + 2k + 14 = 12k^2 + 2k + 14 = 2(6k^2 + k + 7).$$

Now, we know the term $2(6k^2 + k + 7)$ is even by **Def 2.2**, which contradictions our original assumption.

If n is odd, then $n = 2k + 1$ for some integer k. Thus, $3n^2 + n + 14$

$$= 3(2k + 1)^2 + 2k + 15 = 12k^2 + 14k + 18 = 2(6k^2 + 7k + 9).$$

Now, we know the term $2(6k^2 + 7k + 9)$ is even, which contradicts our original assumption.

So, assuming that $\exists n : 3n^2 + n + 14$ is odd always leads to a contradiction. Therefore, $\forall n : 3n^2 + n + 14$ is even.

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